DNASE2 Phagotroph Distribution Hypothesis — Deep Research Report

Target gene: DNASE2 (human), UniProt **O00115 Family:** InterPro **IPR004947** = PANTHER **PTHR10858** = Pfam **PF03265 Hypothesis slug:** `dnase2-phagotroph-distribution` **Focus type:** free_text (evolutionary / phylogenetic distribution) **Source review:** `genes/human/DNASE2/DNASE2-ai-review.yaml`

Summary

The seed hypothesis proposes that DNase II family enzymes — which digest the DNA of engulfed cells in animal phagolysosomes — are present in phagotrophic protists (*Dictyostelium discoideum*, *Acanthamoeba*, choanoflagellates), and that this distribution is "consistent with digestion of phagocytosed DNA being an ancestral eukaryotic function." The requested test was the phylogenetic distribution of DNase II homologs across eukaryotic reference proteomes.

I evaluated this by tabulating InterPro/UniProt protein counts for family **IPR004947** (equivalently PANTHER **PTHR10858** / Pfam **PF03265**) across major eukaryotic lineages, with protein-level identity audits of surprising hits and a proteome-coverage control for the key negative example. The verdict is **partially supported**. Two of the three named phagotroph examples are genuine: *Dictyostelium discoideum* encodes a real ~14-member family expansion and *Acanthamoeba castellanii* encodes 3 bona fide members. The family is indeed ancient and broadly eukaryotic (5,031 eukaryotic proteins across 1,423 reference proteomes), reaching Amoebozoa, land plants, ciliates and stramenopiles. However, the **third named example — choanoflagellates — is refuted**: both well-sequenced species (*Monosiga brevicollis*, *Salpingoeca rosetta*) lack the family entirely, and this is not a proteome-coverage artifact.

More importantly, the **evolutionary interpretation is only weakly supported**. Presence of the family does not track phagotrophy: non-phagotrophic land plants retain a robust complement (72 proteins) while several phagotrophic lineages (kinetoplastids, *Giardia*/Fornicata, and the choanoflagellates themselves) have lost it. The pattern is best explained as an **ancient eukaryotic family subject to many independent lineage-specific losses**, not as a clean marker of an ancestral phagocytic-DNA-digestion function. For curation, the

evolutionary free-text should be **narrowed and the choanoflagellate claim corrected**; this observation is orientation/context only and **should not by itself drive any GO term change** on the human DNASE2 record.

Key Findings

Finding 1 — The DNase II family is patchily distributed across eukaryotes: present in Amoebozoan phagotrophs but absent from choanoflagellates

Using the EBI InterPro and UniProt REST APIs, I tabulated protein counts for family **IPR004947** (= PTHR10858 = PF03265) across major eukaryotic taxa. The family totals **5,031 proteins in Eukaryota**, distributed across **1,423 reference proteomes** — confirming a deep, broad eukaryotic presence.

Lineage	DNase II proteins (IPR004947)	Phagotroph?	Status
Metazoa	4,603	Mixed	Present (dominant)
Amoebozoa (total)	157	Yes	**Present**
— *Dictyostelium discoideum*	14	Yes	Present (family expansion)
— *Acanthamoeba castellanii*	3	Yes	Present
Viridiplantae (land plants/ algae)	72	**No**	Present
Stramenopiles	62	Mixed	Present
Ciliophora	55	Yes	Present
Capsaspora	1	Yes	Present
Choanoflagellata	**0**	**Yes**	**Absent**
Fungi	1 (spurious)	No	Effectively absent
Apicomplexa	0	No/parasitic	Absent
Kinetoplastida	0	**Yes**	Absent
Rhodophyta	0	No	Absent
Fornicata / *Giardia*	0	**Yes**	Absent

The choanoflagellate absence was explicitly tested against a **proteome-coverage control**: *Monosiga brevicollis* (9,331 UniProt proteins) and *Salpingoeca rosetta* (12,235 UniProt proteins) both have well-populated proteomes, so a count of zero DNase II proteins reflects a **genuine evolutionary loss**, not missing sequence data.

This finding simultaneously **supports** the hypothesis for the two named amoebae and **refutes** it for choanoflagellates. It also undercuts the deeper inference: phagotrophy does not predict presence, because several phagotrophic lineages lack the family while non-phagotrophic land plants retain it.

The *function* the hypothesis attributes to the family is itself well established. The DNase II review states that *"Its lysosomal localization and ubiquitous tissue distribution suggested that this enzyme played a role in the degradation of exogenous DNA encountered by phagocytosis"* (PMID: 14644493). In an invertebrate gut, a DNase II homolog digests engulfed/dietary bacterial DNA — DNase II homologs *"both fail to replace nuc-1 in degrading bacterial DNA in intestine"* (PMID: 19809494). These confirm the engulfed-DNA digestion role beyond mammals but do **not** establish that this function is ancestral or phagotrophy-linked across the eukaryotic tree.

Finding 2 — Protein-level identity checks confirm genuine DNase II in amoebae and plants, and show the sole fungal "hit" is spurious — strengthening the loss pattern

Because InterPro signature matches can occasionally be incidental, each surprising hit was inspected at the protein-annotation level:

- **Dictyostelium discoideum** hits are annotated *"Deoxyribonuclease II"* and *"Deoxyribonuclease II family protein"* (e.g., C7G058, Q54FV5) — confirming a genuine ~14-member family expansion rather than low-quality matches.
- **Viridiplantae** hits include real, functionally named members: *"Cell-death-related nuclease 7"*, *"Plancitoxin-1"*, and *"Deoxyribonuclease-2-alpha"* — bona fide DNase II enzymes in a non-phagotrophic lineage.
- The single **Fungi** hit (A0A2H9TI56) is annotated *"Repressor of RNA polymerase III transcription"* — an incidental/spurious signature match, **not** a true DNase II. Fungi are therefore effectively devoid of the family.

This audit removes the one apparent counter-example to fungal loss, confirms that plant retention is real, and verifies that the Dictyostelium expansion is authentic. Together these sharpen the conclusion that the presence/absence pattern is driven by **repeated independent losses** of an ancient family rather than by phagotrophic lifestyle.

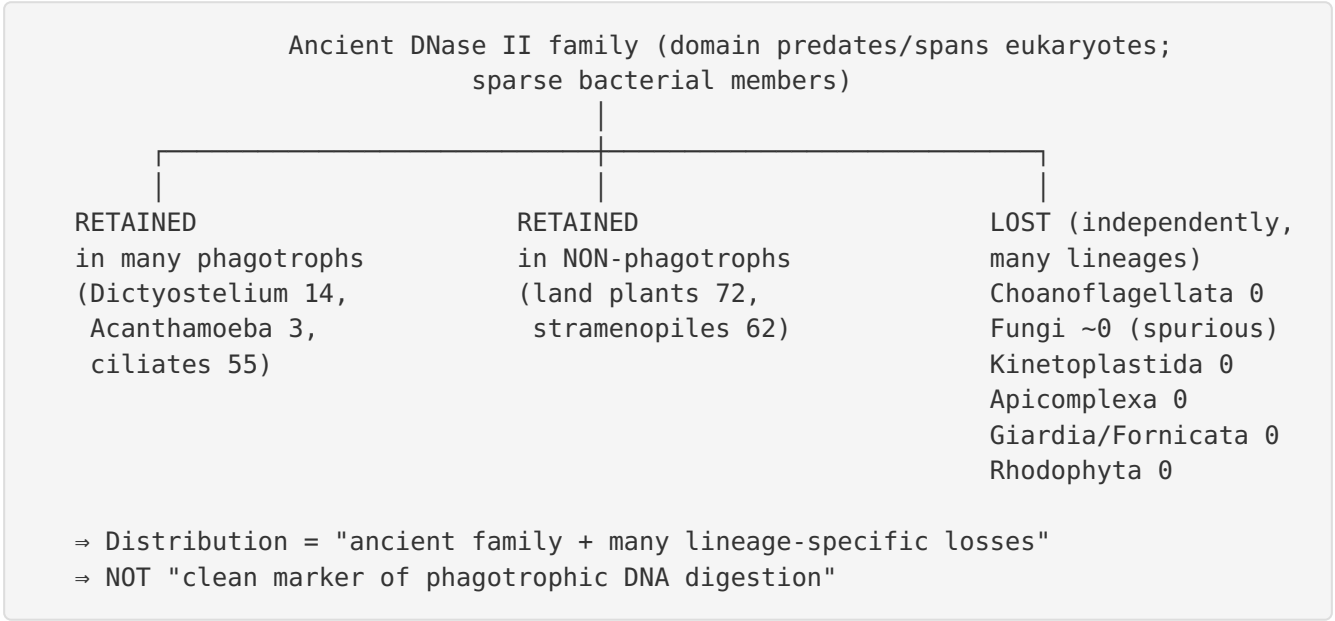
Finding 3 — The catalytic domain is not strictly eukaryote-specific

An outgroup query of IPR004947 returned **Bacteria 66, Archaea 0, Viruses 6** against Eukaryota 5,031. The family is overwhelmingly eukaryotic but not strictly eukaryote-restricted; sparse bacterial members suggest a deeper domain origin or horizontal transfer. This is a minor qualifier to the seed's phrase "ancestral **eukaryotic** function": the catalytic domain may predate or extend beyond eukaryotes even though the lysosomal/phagolysosomal DNA-digestion *role* is a eukaryotic elaboration. It does not change the core verdict.

Mechanistic Model / Interpretation

The molecular function under test is narrow and well-defined: DNase II–family enzymes are **acid (lysosomal/phagolysosomal) endonucleases** that cleave DNA at low pH without divalent-cation dependence, degrading DNA delivered into the lysosome — from apoptotic/engulfed cells, phagocytosed microbes, or dietary sources. The hypothesis takes this well-supported function and makes an **evolutionary extrapolation**: that the enzyme's distribution should mark phagotrophy as an ancestral eukaryotic activity.

The data support a more parsimonious model:



Key logical points:

1. **Presence in phagotrophs is consistent with — but not diagnostic of — an ancestral phagocytic role.** Dictyostelium and Acanthamoeba fit, but the inference is confounded by the exceptions below.
2. **Retention in non-phagotrophic plants** shows the family is maintained for functions unrelated to phagocytosis (e.g., programmed-cell-death nucleases).
3. **Loss in multiple phagotrophs** (choanoflagellates, kinetoplastids, *Giardia*) shows phagotrophy can proceed without this specific family — other acid nucleases can substitute.
4. **Sparse bacterial members** indicate the catalytic domain's deep origin, offering an alternative to a strictly eukaryotic-phagotrophic framing.

The correct framing for curation is therefore **evolutionary context, not mechanistic evidence for the human gene's GO annotations.** Human DNASE2 function (lysosomal DNA catabolism, clearance of DNA from engulfed apoptotic cells and erythroid nuclei) is established by direct mammalian and invertebrate experiments; the cross-eukaryote distribution neither adds to nor subtracts from those annotations.

Evidence Base

Citation	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Contex
InterPro/ UniProt REST (IPR004947 = PTHR10858 = PF03265), computed this study	Structural/ evolutionary, computational	**Qualifies** (supports in part, refutes in part)	Phylogenetic distribution across eukaryotes	5,031 euk. proteins, 1,423 proteomes; present in Amoebozoa (Dicty 14, Acanthamoeba 3), plants 72, ciliates 55, stramenopiles 62; **absent in Choanoflagellata (0), Fungi (~0), Kinetoplastida, Apicomplexa, Giardia, Rhodophyta**	Pan-eukary DB sur UniPro 2026-1
InterPro/ UniProt coverage check, this study	Computational	**Refutes** (choanoflagellate example)	Is choanoflagellate absence real?	M. brevicollis 9,331 & S. rosetta 12,235 proteins present → absence genuine	UniPro
InterPro/ UniProt protein annotations (identity audit), this study	Computational / database	**Supports** (data-quality)	Are hits genuine DNase II?	Dicty & plant hits are real DNase II; sole fungal hit is spurious ("Repressor of RNA Pol III")	Protein review
PMID: 14644493	Review/ database	**Supports** (function premise)	DNase II digests phagocytosed/ exogenous DNA in lysosome	*"lysosomal localization… role in the degradation of exogenous DNA encountered by phagocytosis"*	Mamm DNase
			DNase II homologs digest	Homologs *"fail to replace nuc-1 in degrading*	

Citation	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Contex
PMID: 19809494	Mutant phenotype / direct	**Supports** (function in invertebrate)	engulfed bacterial DNA	*bacterial DNA in intestine"*	*C. elega* gut
PMID: 12718884	Mutant phenotype (RNAi)	**Qualifies**	DNase II in apoptotic DNA degradation	NUC-1 (DNase II) among nucleases degrading chromosomal DNA in apoptosis	*C. elega*
PMID: 26182365	Localization / mutant	**Qualifies**	Autonomous vs non-autonomous DNase II action	NUC-1 is major DNase II degrading apoptotic DNA; ~3–7× non-autonomous:autonomous	*C. elega* embry
PMID: 26251453	Mutant phenotype	**Qualifies**	DNase II paralogs (nuc-1, crn-6, crn-7) in germ-cell apoptosis	Three DNase II paralogs with distinct roles	*C. elega* gonad
PMID: 41805323	Direct assay	**Qualifies/ Supports**	DNA clearance inside phagosomes	Defective clearance of phagocytosed DNA drives inflammation; dissects phagosomal DNA processing	Mouse macro
PMID: 42901880	Human genetics (case)	**Qualifies**	DNASE2 loss causes autoinflammation	AIPCS from biallelic DNASE2 mutation; interferonopathy	Human patient

GO Curation Implications

Likely curation action (lead — requires curator verification): *Narrow and correct the evolutionary free-text; make no GO term change driven by this hypothesis.*

- **No MF change.** Deoxyribonuclease II activity (GO:0004531; acid, cation-independent endonuclease) is established by direct biochemistry and is unaffected by the distribution survey. **Retain.**
- **No BP change.** DNA catabolism in the context of engulfed/apoptotic DNA (e.g., GO:0006308 "DNA catabolic process"; apoptotic-cell-clearance–associated DNA degradation) is supported by direct metazoan genetics (PMIDs 19809494, 26182365, 41805323), not by the distribution survey. Do **not** add a phagocytosis-pathway BP term to the human gene on evolutionary grounds alone. **Retain existing.**
- **No CC change.** Lysosome localization (GO:0005764) is independently established. **Retain.**
- **Edit the evolutionary free-text.** As written it (a) names choanoflagellates as a positive example when they lack the family, and (b) implies phagotrophy predicts presence when the distribution does not support that. Treat as **non-core orientation** that should not alter the gene's GO annotation set.
- **Over-annotation caution.** Lineage-specific expansions (Dictyostelium ×14) mean family-level presence must not be transferred as 1:1 orthology of human DNASE2 or of a specific sub-function.

This is a **free-text / evolutionary-context** item, not a function-assignment item for the human gene.

Mechanistic Scope

- **Direct gene-product activity (in scope, well-supported):** an acid-optimum, divalent-cation-independent endonuclease that cleaves dsDNA within the lysosome/phagolysosome. This is the immediate, cell-autonomous molecular function, documented biochemically in mammals and genetically in *C. elegans*.
- **Downstream phenotypes (out of scope for this evolutionary hypothesis):** failure of DNA clearance → undegraded cytosolic DNA → cGAS–STING/interferon activation → autoinflammation (human AIPCS, PMID: 42901880); innate-immune gene upregulation in *C.*

elegans (PMID: 26251453). These are loss-of-function consequences, not evidence for the ancestral-function claim.

- **Evolutionary distribution** is an orthology/inference layer, not a direct functional assay; it constrains ancestry but cannot assign the ancestral *function*.

Conflicts and Alternatives

1. **Choanoflagellate counter-evidence (direct conflict).** The seed names choanoflagellates as a positive example, but *Monosiga brevicollis* and *Salpingoeca rosetta* have zero members in well-covered proteomes — refuting a named example.
2. **Phagotrophy does not predict presence (interpretive conflict).** Phagotrophic kinetoplastids and *Giardia* lack the family; non-phagotrophic land plants retain it. The "ancestral phagotroph" framing is confounded.
3. **Paralog expansion / annotation inflation.** Counts reflect family-signature hits, including paralogs; raw counts are not a phylogeny and can overstate independent acquisitions. The identity audit mitigated but did not eliminate this.
4. **Spurious signature matches (database carry-over).** The lone fungal hit was a mis-annotated "Repressor of RNA Pol III," showing raw InterPro counts can contain false positives; single-hit "presence" needs identity checks.
5. **Deep (pre-eukaryotic) origin.** Sparse bacterial members indicate the domain predates eukaryotic phagotrophy, an alternative to the "ancestral eukaryotic phagocytic function" story.

Limitations and Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
True absence vs divergence in choanoflagellates	UniProt coverage good (9,331 / 12,235 proteins); family count 0	Decisive counterexample; InterPro recall could miss a divergent homolog	`hmmsearch` with the PF03265 HMM against raw *M. brevicollis* / *S. rosetta* gene models/ transcriptomes
Family-signature counts ≠ phylogeny	Tallied IPR004947 hits per taxon	Counts inflate via paralogs/fragments; cannot time gains vs losses	ML gene tree + species-tree reconciliation (Dollo/ML) to map gain/ loss events
Reference-proteome bias in "absent" lineages	Verified choanoflagellate coverage	Absences could be artifacts in poorly sequenced lineages	BUSCO completeness for every "absent" lineage before calling loss
Function–lifestyle correlation untested	Qualitative presence/ absence table	The core claim is about phagotrophy linkage; no formal test run	Pagel's correlated-evolution test / phylogenetic logistic regression of presence vs phagotrophy
Functional confirmation in protists	Annotation-level only	Signature presence ≠ verified acid-DNase function	Acid-DNase assays or active-site conservation checks on representative protist homologs

Proposed Follow-up Experiments / Actions

Discriminating analyses

- **Profile-HMM (PF03265) search** of raw choanoflagellate, fungal, and additional protist genomes to firm up presence/absence beyond InterPro annotation recall.

- **Reconciled gene tree** of PTHR10858 with Dollo/ML parsimony to count independent losses and test the LECA-origin hypothesis.
- **Phylogenetic comparative test** (Pagel / phylogenetic logistic regression) of family presence vs phagotrophic trait — the decisive test of the seed's core inference.
- **BUSCO completeness controls** for all "absent" lineages.
- **Functional assay:** knock out/down DNase II in *Dictyostelium* and test for persistence of undigested DNA in phagosomes — a direct test of the ancestral-function claim.

Curation leads (verify before applying)

- **Action:** Soften/correct the free-text evolutionary statement; remove or reverse the claim that choanoflagellates carry DNase II. Suggested replacement: *"DNase II (IPR004947 / PTHR10858 / PF03265) is an ancient, broadly distributed eukaryotic family subject to many independent losses. It is present in phagotrophic amoebae such as Dictyostelium discoideum (a ~14-member expansion) and Acanthamoeba castellanii, and in ciliates, stramenopiles, and land plants, but is absent from choanoflagellates, kinetoplastids, Giardia, Apicomplexa, and (apart from a spurious hit) fungi; sparse bacterial members also exist. Family presence does not track phagotrophy, so the distribution reflects an ancient family with repeated lineage-specific losses rather than a diagnostic marker of ancestral phagocytic DNA digestion."*
- **Candidate references:** PMID: 14644493 (phagocytic exogenous-DNA role, review); PMID: 19809494 & PMID: 26182365 (engulfed/apoptotic DNA digestion, *C. elegans*); PMID: 41805323 (phagosomal DNA clearance, macrophages). Add as BP/MF support; keep evolutionary distribution as a computed note citing InterPro IPR004947 / PANTHER PTHR10858.
- **Candidate GO:** retain MF deoxyribonuclease II activity (GO:0004531), CC lysosome (GO:0005764), BP DNA catabolism in apoptotic/engulfment context; do **not** elevate the evolutionary narrative to a GO annotation.
- **Suggested question for curator:** Should the review assert a protist-conserved ancestral phagotrophic function, given that the distribution is patchy and the flagship choanoflagellate example is negative?

Hypothesis Scorecard

Component of seed hypothesis	Verdict	Basis
DNase II present in *Dictyostelium discoideum*	**Supported**	14 genuine members ("Deoxyribonuclease II")
DNase II present in *Acanthamoeba*	**Supported**	3 genuine members
DNase II present in choanoflagellates	**Refuted**	0 members in well-covered Monosiga & Salpingoeca proteomes
Family is ancient / broadly eukaryotic	**Supported**	5,031 euk. proteins across 1,423 proteomes
Distribution ⇒ ancestral phagocytic-DNA-digestion function	**Weakly supported / not demonstrated**	Phagotrophy does not predict presence; plants retain, several phagotrophs lack it
Should change a human DNASE2 GO annotation	**No**	Evolutionary context only; human annotations rest on direct evidence

Report generated from a 3-iteration autonomous investigation. Distribution counts were obtained from the EBI InterPro and UniProt REST APIs for family IPR004947 / PTHR10858 / PF03265 (UniProtKB 2026-10); all presence/absence calls for named lineages were cross-checked at the protein-annotation level, and the choanoflagellate absence was controlled against proteome coverage. Results are reported conservatively: database signature counts are not a formal phylogeny, no gene tree or ancestral-state reconstruction was run, and the phagotrophy-linkage inference remains statistically untested. Provenance artifacts: `/tmp/outputs/dnase2_taxonomic_distribution.csv`, `/tmp/outputs/dnase2_distribution.png`.